

2-Mark Questions & Answers -- MPCA Unit 1: Overview of Computer Architecture & Organization

Q1. Differentiate between Computer Architecture and Computer Organization.

Feature	Computer Architecture	Computer Organization
Focus	High-level logical aspects visible to	Hardware transparent operational units
Examples	Instruction set, addressing modes, dat	Control signals, memory technology, bu

Q2. What is the Von Neumann bottleneck

The Von Neumann Bottleneck refers to the throughput limitation caused by the shared single system bus between the CPU and memory for both instruction fetches and data transfers. Because instructions and data cannot be accessed simultaneously over the same bus, the CPU is forced to wait (CPU stall), constraining overall processing speed.

Q3. List any four performance measures of a computer system.

1. Clock Cycle Time / Frequency (GHz)

2. Ins

Q4. Define MIPS and FLOPS.

* MIPS (Million Instructions Per Second): A measure of CPU processing speed defined as:

MIPS

* FLOPS (Floating-Point Operations Per Second): A metric measuring hardware performance in numerical, scientific, and AI floating-point calculations (GFLOPS, TFLOPS, PFLOPS).

Q5. What is meant by instruction set architecture (ISA)

Instruction Set Architecture (ISA) is the abstract boundary/interface between computer hardware and software. It defines everything a programmer or compiler needs to know to execute code on a CPU, including supported instructions, data types, register sets, memory addressing modes, and interrupt handling models.